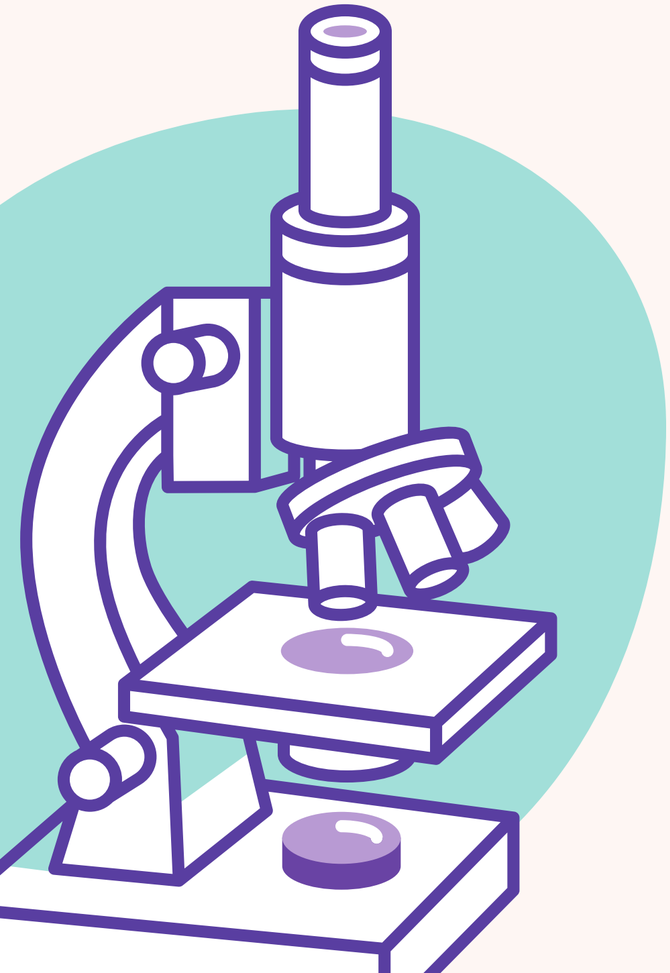
Chemistry in Everyday Life





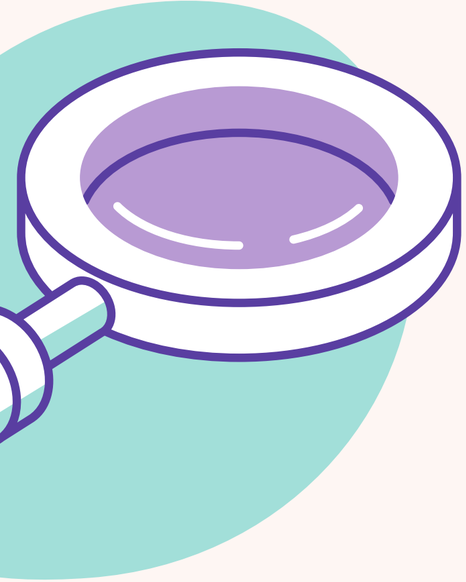
What are drugs?

Drugs are chemicals of low molecular masses (~100-500u). These drugs interact with macro molecular targets and produce a biological response.



When are drugs called medicine?

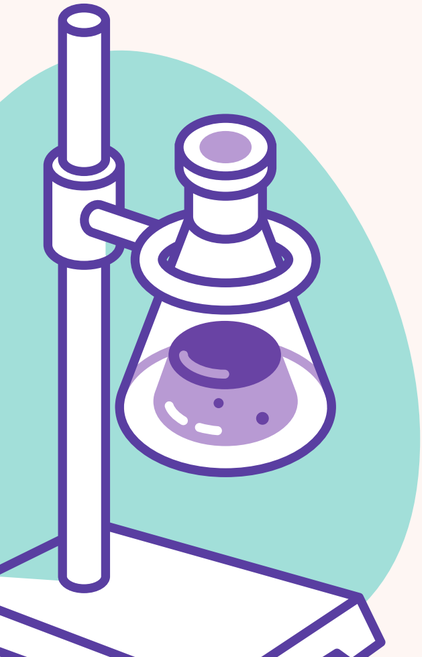
The drugs that produce a therapeutic biological response and used in diagnosis, prevention and curing of diseases are called medicines



Macro molecules such as nucleic acids, proteins, carbohydrates and lipids are called drug targets.

Proteins which perform the role of biological catalysts in the body are called enzymes. Proteins that are crucial to communcation system in the body are called receptors.

Sites different from active site of enzyme where a molecule can bind an effect the active site is called allosteric site.





Chemicals that remove the access acid in the stomach and maintain the ph to normal level are called antacids.
eg. Omeprazole, Lansoprazole.



tranquilizers are drugs mainly used in the management or treatment of psychoses and neuroses
eg. Luminal, Seconal, Barbituric Acid



Analgesics reduce or abolish pain without causing imparement of consciousness, mental confusion paralysis aur some other disturbance of nervous system. Classified into non narcotic and narcotic.

Non narcotic drugs are analgesics but they have no addictive properties.
eg. Ibuprofen, Aspirin.



Antibiotics are chemical substances produced by microorganisms that inhibit the growth or destroy microorganisms.
eg. Penicillin, Chloramphenicol



Antiseptics are chemical compounds that kill or prevent the growth of microorganisms
eg. Dettol, Tincture of Iodine



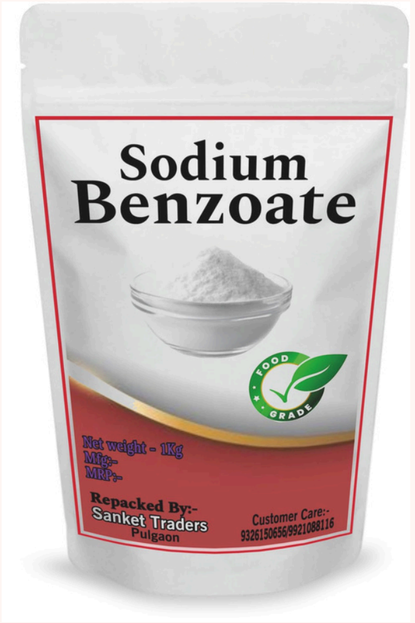
Disinfectants are chemical compounds used for killing or preventing the growth of micro these are applied on inanimate objects like drainage systems.
eg. 4% aq. solution of formaldehyde



Artificial sweetners are chemical substances which are sweet in taste but do not add any calories to our body.
eg. Saccharin, Aspartame



Chemical substances which are used to protect food against bacteria yeast and moulds are called preservatives.
eg. Sodium metabisulpahte, Sodium benzoate.



butylated hydroxy toluene (BHT) and butylated hydroxy anisole (BHA) are antioxidants.